Fiscal Year:	FY 2005	**Task Last Updated:**	FY 08/31/2005
PI Name:	Dingfelder, Michael Ph.D.		
Project Title:	Patterns of Energy Deposition by HZE Particles in Cellular Targets		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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PI Email:	dingfelderm@ecu.edu	**Fax:**	FY 252-328-6314
PI Organization Type:	UNIVERSITY	**Phone:**	252-328-0882
Organization Name:	East Carolina University		
PI Address 1:	Department of Physics		
PI Address 2:	Mailstop 563		
PI Web Page:			
City:	Greenville	**State:**	NC
Zip Code:	27858	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Toburen, Larry H (East Carolina University)		
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The interpretation of data from mechanistic studies on radiation biology requires detailed descriptions of the local patterns of energy deposition in the appropriate cellular or sub-cellular sites to explore the effects of radiation quality on biological response. The objective of this project is to provide Monte Carlo (MC) simulations on the local patterns of energy deposition by HZE particles in the appropriate cellular environment including effects of micro-, and macro-geometrical considerations. The calculations focus on interpretation of microscopic “event” or “dose” distributions appropriate to recently initiated studies of HZE radiation induced leukemia in animal and cellular models [NSCOR Program, Colorado State University, R. Ulrich, PI] and those encountered in the production of clustered damage in DNA [e.g., Brookhaven National Laboratory, B Sutherland, PI]. The progenitor cells for leukemia are haematopoietic stem cells that are located in clusters within marrow cavities having dimensions on the order of a few hundred microns. These cavities are surrounded by trabecular bone matrix, which forms a wall of several hundred

Task Description:	microns thickness around the marrow. The dimensions and geometry of the bone marrow are too irregular for direct dosimetry measurements and dose-averaging methods are inappropriate for chronic low dose-rate (single particle) exposures. Detailed Monte Carlo charged particle track structure simulation will provide quantitative information on spatial patterns of energy deposition in cellular targets of dimensions as small as a few nanometers on a particle-by-particle basis; these distributions are the starting point to characterize the detailed nature of radiation-induced damage produced by high-LET radiation. The modeling approach will combine the capabilities of full event-by-event Monte Carlo track simulation codes with codes employing condensed history capabilities; the latter is more appropriate for the description of high-energy secondary electrons. The event-by-event ion code PROTRAC (which is part of the biophysical simulation tool PARTRAC) developed at GSF Neuherberg for protons and alpha particles will be extended for heavy charged particle (HZE) interactions using appropriate relativistic theory. Generated secondary electrons that traverse specific regions of interest, e.g., bone marrow cells or their nuclei, will be followed also event-by-event using ETRAC (the electron transport code of PARTRAC) whereas electron transport in adjoining tissues and the surrounding structural material will be followed by the more computer efficient condensed history general purpose MC transport code PENELOPE developed at the University of Barcelona. Appropriate interaction cross sections for a wide range of target materials will be added to the ETRAC and PROTRAC codes, specifically to explore the effects of bone-tissue interfaces on energy deposition in bone marrow cells. Comparison of the spatial pattern of energy deposited in volumes of interest obtained from models of the HZE particles with that obtained from low-LET radiation will provide a firm basis to interpret radiation quality effects on biological response and the mathematical structure developed is amenable to scaling to volumes appropriate to human dimensions for considerations of risk assessment. The objective of this project is to provide Monte Carlo (MC) simulations on the local patterns of energy deposition by HZE particles in the appropriate cellular environment including effects of micro-, and macro-geometrical considerations. The focus will be on Leukemia, i.e., on the bone/bone-marrow interface from realistic mouse or human dimensions down to single cells. Little is known on the interaction of HZE particles with the human body. HZE particles are part of the Galactic Cosmic Rays (GCR) which exist in deep space and cannot be shielded completely inside spacecrafts. The latency time of leukemia, 2-3 years, coincides more or less with the mission time for the planned expedition to Mars.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The project was iniciated with the start of the Fall semester 2004. We succeeded in recruting two PhD students for the project during the year and have started work in two directions: First, we have started the developement of a module for the track structure simulation code PARTRAC to simulate HZE particle tracks in liquid water and calcium (as bone surrogate). Second, we have started work on the development of a realistic bone/bone-marrow target suitable for use with Monte Carlo codes. Regarding the HZE particle transport we recently have presented a fully relativistic description of the plane-wave Born approximation for inelastic scattering in condensed matter (J.M. Fernandez-Varea, F. Salvat, M. Dingfelder, D. Liljequist, Nucl. Instr. Meth. B 229 (2005) 187-218.) which includes also relativistic medium effects like the Fermi density effect in a consistent way. The Fermi density effect is a relativistic polarization effect and becomes more and more important with increasing energies. The (relativistic) plane-wave Born approximation relates the (macroscopic) dielectric response function of the target with the (microscopic) interaction cross sections. Within this theory, interaction cross sections for bare heavy ions (HZE particles) are obtained from proton interaction cross sections by scaling laws. We have updated our recent model of the dielectric response function of liquid water (Dingfelder et al., Radiat. Phys. Chem. 53 (1998) 1-18.) since new pertinent experimental information became available (Hayashi et al., Proc. Natl. Acad. Sci. 97 (2000) 6264-6266.). We also started modeling the dielectric response function of calcium. Currently we are calculating the interaction cross sections for protons and HZE particles in liquid water at relativistic energies. We initiated a study to evaluate the effects of using surrogate materials for bone and bone marrow in our simulations. For that purpose we modeled a simple interface consisting of two different materials (cortical bone and blood (ICRU definitions) as the ideal case, and calcium and water as surrogates) and calculated depth dose distributions for different y-ray (Co-60 and Cs-137) and x-ray qualities using the general purpose Monte Carlo code PENELOPE. PENELOPE simulates coupled electron-photon transport in all atoms and ICRU defined materials. We learn that water is a good surrogate for blood while a calcium surrogate produces much higher dose perturbation for Co-60 radiation. We have repeated the simulations using 4 MeV electrons instead of gamma rays. Also in this case we see slightly higher dose perturbations when using calcium instead of cortical bone. As a summary, we may use water as a substitute for blood and calcium for cortical bone, but should be aware of the limitations of this substitution. We also started work on a realistic but practical bone/bone-marrow interface for use in Monte Carlo simulations. For that purpose we use the geometrical package of PENELOPE. Our recent model is based on the following assumptions: (1) the bone is spherical. (2) Bone marrow is enclosed in spheres, randomly distributed inside the bone. (3) The bone-marrow sphere diameters have the same distribution as measured chord lengths in (human) trabecular bones. (4) Known ratios of bone to bone-marrow volume and surface are used to determine the number of bone-marrow cavities. Parameters and chord length distributions are obtained from the literature.
Bibliography Type:	Description: (Last Updated: 11/09/2018)
Abstracts for Journals and Proceedings	Toburen, L.H.; Dingfelder, M.; Gersh, J.A.; Lapicki, G.; Christou, C.I.; Conticchio, C.D.; Shinpaugh, J.L. "Modeling energy deposition by ionizing radiation in a cellular environment" Department of Energy Low Dose Radiation Research Program Workshop V. Bethesda, MD, April 25-27, 2005. Book of Abstracts, p. 45. , Apr-2005
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Presentation	Dingfelder, M. "Simulation of DNA damage induced by ionizing radiation" Invited talk (Colloquium) given at the Department of Biology, East Carolina University. Greenville, NC, February 17, 2005. Feb-2005
Presentation	Dingfelder, M. "Biophysical charged particle track structure simulations: Models and basic data" Invited talk (Colloquium) given at the Department of Physics, East Carolina University. Greenville, NC, February 11, 2005. Feb-2005
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Presentation	Dingfelder, M.; Toburen, L.H.; Paretzke, H.G. "Ionization cross sections of partially dressed helium ions in liquid water." Talk presented at the Monte Carlo 2005 Topical Meeting: The Monte Carlo method: Versatility unbounded in a dynamic computing world. Chattanooga, TN, April 17-21, 2005. Apr-2005